

Review

Synthesis of fluorinated analogs of natural porphyrins potentially useful for the diagnosis and therapy of cancer

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Abstract

We have synthesized a number of fluorinated analogs of naturally important porphyrin derivatives, such as protoporphyrin and hematoporphyrin, which would be useful for the diagnosis and therapy of cancer, starting from deuteroporphyrin or using ring-closure methodology. Fluorovinyl derivatives of protoporphyrin were synthesized using the Wittig reaction or by employing organozinc reagents. The introduction of trifluorohydroxyethyl groups into deuteroporphyrin gave fluorinated analogs of hematoporphyrin, some of which were resolved to chiral isomers. Ring-closure of pyrrole derivatives carrying trifluorohydroxyethyl groups gave much better yields of chiral fluorinated analogs of hematoporphyrin than the resolution method. This review presents our syntheses of these fluorinated analogs and provides information on localizations of these porphyrins in some tumor cells and tissues. © 2001 Elsevier Science B.V. All rights reserved.

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1. Introduction

Planar macrocycles comprising four pyrrole rings joined by four methine bridges are generally called porphyrin. The most important porphyrin is protoporphyrin (**1**), which is a constituent of hemoglobin and other important enzymes. Porphyrins are highly conjugated, and their main absorption bands have very high extinction coefficients; the intense ‘Soret’ band, found around 400 nm, is a characteristic of this macrocyclic conjugation [1]. Furthermore, porphyrins show a characteristic red fluorescence on irradiation with ultra-violet light. Based on their structural and spectroscopic properties, these porphyrin derivatives are being investigated for application in various fields, one of the recent studies being centered on the use of ‘hematoporphyrin derivatives’ (HpD) to diagnose and treat tumors [2].

It is well known that some porphyrin derivatives can become localized in tumor tissues [3], and especially HpD, obtained by treatment of hematoporphyrin (**2**) with sulfuric acid in acetic acid, is reported to do so efficiently. After administration of HpD, a tumor tissue taking it up fluoresces a reddish color on photoirradiation with laser light, so the early stage of a cancer can be detected. Furthermore, it is known that some porphyrins produce

‘active’ oxygen by photosensitization and hence destroy most cancer cells. This treatment of cancers by irradiation is called photodynamic therapy (PDT). However, HpD used for PDT is a complex mixture of several porphyrins, such as hematoporphyrin (HP; **2**), hydroxyethylvinyldeuteroporphyrin (HVD; **3**), protoporphyrin (PP; **1**), hematoporphyrin diacetate (HDA), and so on [4] (structures of some of these porphyrins are shown in Fig. 1). Therefore, HpD has some drawbacks where clinical use is concerned: namely its low purity and unstable photosensitivity. Recently, Photofrin II was developed as a low-photosensitivity PDT drug. However, it is thought that this drug consists mainly of dihematoporphyrin ester or ether, but the composition is not constant, therefore ambiguity exists concerning which component is effective agent [5]. Further, it is not clear which component of the mixture localizes to cancer cells. In the search for a more effective photosensitizer than Photofrin II, the synthesis and applications of phthalocyanines, chlorins [6], hematoporphyrin oligomers (HPO) [7], Ga-complexes of porphyrin dyes [8], and pheophorbide derivative (PH-1126) [9,10] are being actively investigated.

Absorption of photoenergy by a ground state (S_0) photosensitizer molecule promotes it to an excited singlet state ($^1S^*$), which can undergo conversion to a triplet state ($^3S^*$) via intersystem crossing (ISC). The energy of this triplet state is transferred to ground state oxygen (3O_2) and excited singlet oxygen ($^1O_2^*$) is produced. The excited singlet

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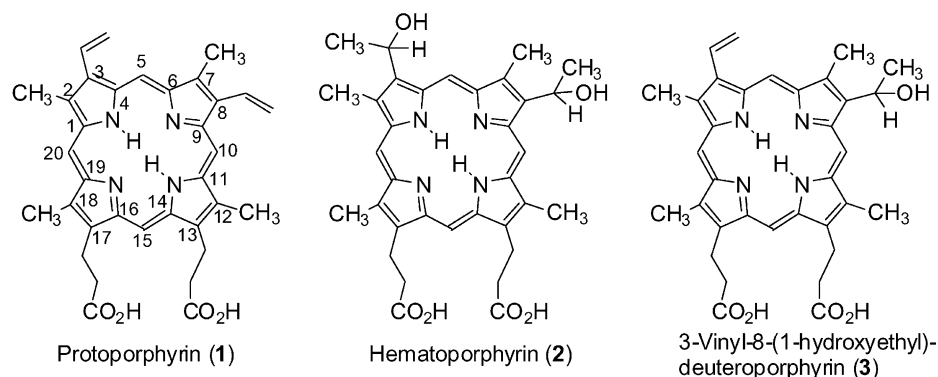
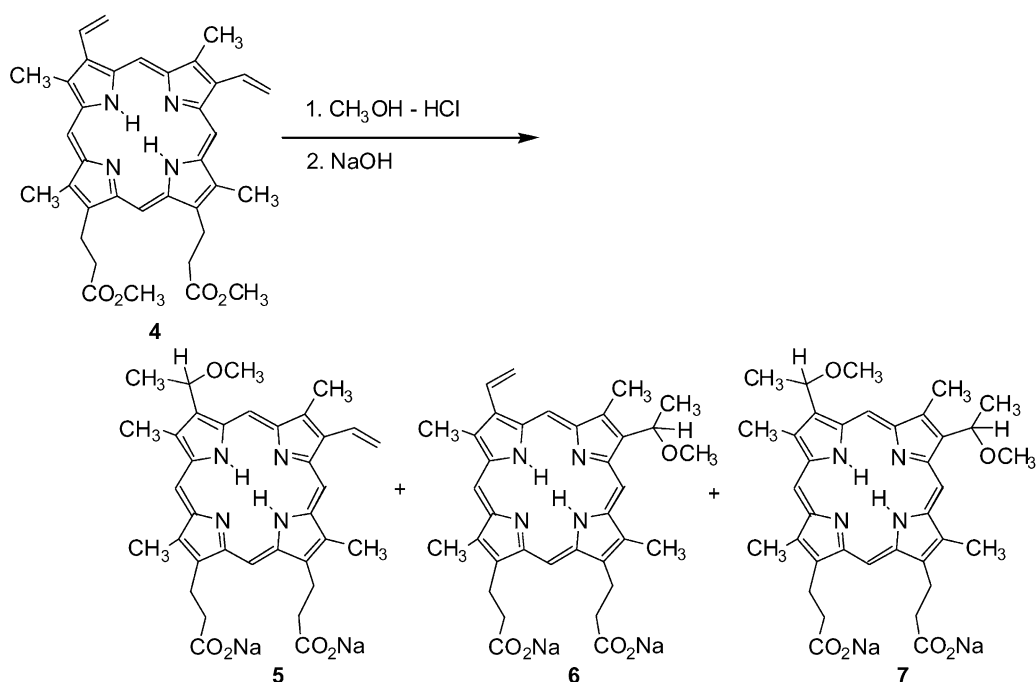


Fig. 1. Some well known porphyrins.

oxygen is highly oxidative and reactive compared with ground state oxygen, so it is called “active” oxygen. Other forms of active oxygen produced are superoxide, hydroxyl radical, hydrogen peroxide, and so on. These active species are produced in cancer tissues by photosensitization of localized porphyrins, and oxidize important enzymes found in cancer cells. As a result, the cells are deactivated and then cancer tissue is necrotized. In the photosensitized reaction by using porphyrin derivatives [11], it is believed that excited singlet oxygen is formed at primary stage. However, contribution of radical forms of active oxygen cannot be eliminated [12]. An excited photosensitizer returns to its ground state after transfer of its triplet state energy to ground-state oxygen. It is then reexcited by further irradiation, so the above cycle of steps repeats until photoirradiation is discontinued.

As mentioned above, Photofrin II used at present for PDT is a complex mixture [5] and not pure, hence alternatives are

being sought. What is needed is a pure porphyrin derivative, which has a high selectivity for tumorous tissue and localizes to specific tumor cells. If discovered, such compound will be very useful for the diagnosis and therapy of cancers, and PDT will develop greatly to allow a specific therapy for each type of cancer. In this connection, we reported previously [13] that treatment of protoporphyrin dimethyl ester (4) with HCl gas in CH₃OH, followed by alkaline hydrolysis, gave three methanol adducts of 1, 3-(1-methoxyethyl)-8-vinyldeuteroporphyrin sodium salt (5), 8-(1-methoxyethyl)-3-vinyl-deuteroporphyrin sodium salt (6) and 3,8-bis(1-methoxyethyl)deuteroporphyrin sodium salt (7), as shown in Scheme 1. Investigation of the localization of each CH₃OH-adduct to human gastric cancer (GCA-1) showed that 5 had an interesting physiological property. Thus, 5 localized specifically to gastric cancer but not to human hepatocellular carcinoma (HCC-1), while 6 and 7 localized little to both types of cancer cell. Compound 7



Scheme 1. Reaction of protoporphyrin dimethyl ester with methanol in the presence of HCl.

showed a similar photodynamic effect to JTC-16 cancer cells on laser irradiation as HpD [14]. These results suggested that some porphyrins might localize selectively to a specific tumor tissue and could be a specific sensitizer for a special kind of tumor. Since that work, we have been investigating the synthesis of fluorinated analogs of porphyrin derivatives which will localize selectively to a specific tumor tissue and a potentially useful for cancer diagnosis and therapy.

The reasons why we chose to study fluorinated analogs are: 1. A fluorine atom is not much larger than a hydrogen atom, hence the incorporation of a fluorine substituent would not change the steric characteristic of the original porphyrin and thereby interfere with its take-up by tumorous tissue. 2. Although carbon-fluorine compounds are often viewed as very stable, some are very reactive, and these reactivities are believed to lead to important biological effects [15,16]. Some of the fluorinated analogs were expected to react with biological components and to show anti-tumor effects. 3. Organic fluorine compounds are quite rare in biological systems. Therefore, if a fluorinated porphyrin localized in tumor tissues, it could be detected by ^{19}F -NMR imaging. The present review deals with our work on the synthesis of fluorinated analogs of natural porphyrin and provides details of the preliminary biological tests carried out.

2. Synthesis of fluorine derivatives of protoporphyrin

2.1. Fluorinated analogs of protoporphyrin with difluorovinyl group(s)

Protoporphyrin (**1**) is a demetallated porphyrin of protoheme, which is a component of hemoglobin. Its zinc complex is used as a medicine, so **1** is important as a bioactive compound, and we decided to tackle the synthesis of some fluorinated analogs. As mentioned above [14], the activity of the CH_3OH -adduct of **1** depends on the site of addition, so that we thought that the vinyl groups at the 3- and 8-positions of protoporphyrin play an important role. Therefore, first we investigated the introduction of fluorine at the vinyl groups (see Scheme 2).

Protoporphyrin dimethyl ester (**4**) obtained from protohemin was photooxidized by air with white light to give so-called photoporphyrins (**8** and **9**), both isomers of which were separated by column chromatography. These compounds were reduced with sodium borohydride, followed by rearrangement of the diols with acid, to glycols (**10** and **13**). The glycols were cleaved with periodic acid to give 3-formyl-8-vinyl- and 8-formyl-3-vinyl-deuteroporphyrin dimethyl esters (**11** and **14**) according to the literature [17] with some modifications [18,19], since the yields of these formyl derivatives (**11** and **14**) were much lower than reported. Thus, the reduction of photoporphyrins (**8** and **9**) with NaBH_4 was carried out at room temperature, since tarry substances were formed in fairly large amounts at a higher

temperature [17]. After acidification and extraction with CH_2Cl_2 , the glycols (**10** and **13**) were each obtained in 80% yields after purification by column chromatography. These glycols were oxidized with $\text{HIO}_4\cdot 2\text{H}_2\text{O}$ in dioxane instead of using $\text{NaIO}_4\text{--H}_2\text{SO}_4$ in benzene- CH_2Cl_2 [17], and the desired formyl derivatives (**11** and **14**) were obtained in 80% yields [18,19]. The bis-formyl derivative **17** was obtained by OsO_4 -oxidation of protoporphyrin dimethyl ester (**4**), followed by NaIO_4 -oxidation [20]. These formyl derivatives (**11**, **14** and **17**) were treated with sodium chlorodifluoroacetate in the presence of triphenylphosphine in *N*-methylpyrrolidone (NMP) to give difluoro- and tetrafluoroporphyrin derivatives (**12**, **15** and **18**) (see Scheme 2). Use of NMP as a solvent was crucial. Diglyme, which is commonly used for this reaction, did not give a good result.

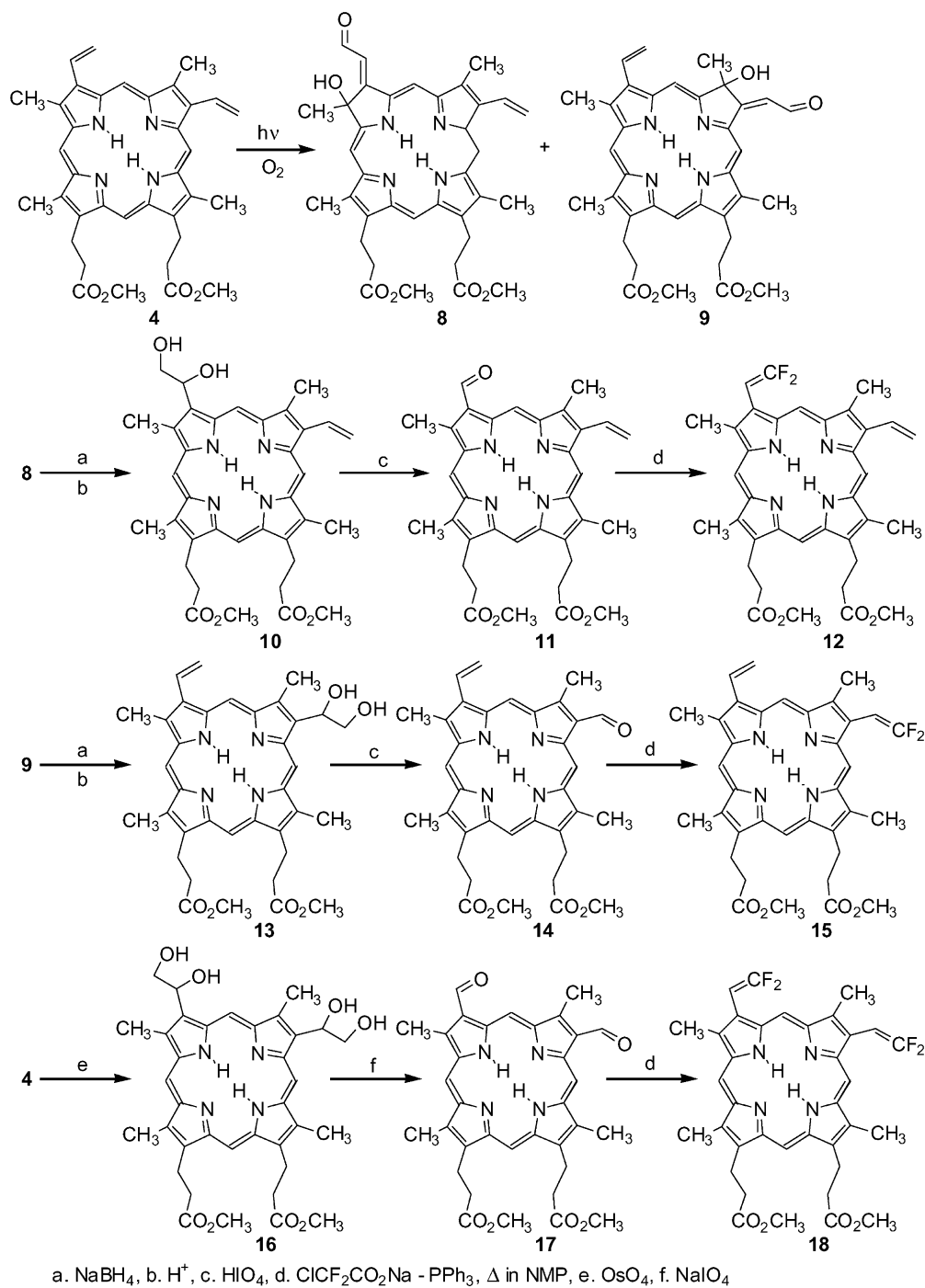
The difluorovinyl derivatives **12**, **15** and **18** were hydrolyzed with NaOH in toluene- CH_3OH to give the corresponding sodium salts (**12'**, **15'** and **18'**), and each sodium salt was subjected to the preliminary test of uptake by human gastric cancer (MKN-45). The results are shown in Fig. 2. This figure shows that 8-FPPN (**15'**) has a high localizability to cancer cells and stomach, being localized specifically to gastric cancer cells. When these compounds were tested using rat ascite hepatoma cells, 3,8-FPPN (**18'**) was taken up the most efficiently [18,19]. These results supported our expectation that one porphyrin could be taken up selectively by one cancer and another porphyrin by another cancer. Thus, we planned to synthesize and test other fluorinated analogs of protoporphyrin.

2.2. Synthesis of fluorinated analogs of protoporphyrin containing trihalovinyl groups using reactive fluorovinylzinc reagents

Since analogs of protoporphyrins containing one (**12**, **15**) or two (**18**) difluorovinyl groups in the place of vinyl groups showed an interesting behavior towards cancer (Section 2.1), we decided to synthesise trifluorovinyl counterparts by the reaction of iodoporphyrin with trifluorovinylzinc chloride generated as reported by Normant's group [21]. At first, we followed their procedure for making the reagent, but we obtained the desired product only in a poor yield. Modification of the procedure gave new fluorovinylzinc reagents which were much more reactive than the reported one, and the first part of this section deals with our new methodology. Applications to the synthesis of various fluorinated analogs of protoporphyrin are presented in Section 2.2.2.

2.2.1. New procedure for the synthesis of fluorovinylzinc reagents and their application for the synthesis of bis(trifluorovinyl)- and bis(1-chloro-2,2-difluorovinyl)-deuteroporphyrin dimethyl esters [22]

Normant's group effected the synthesis of trifluorovinylzinc chloride (**21**) by the reaction of chlorotrifluoroethene (**19**) with *sec*-butyllithium at -60°C , followed by the addition of zinc chloride at -100°C , and treated aryl iodides with



Scheme 2. Derivatization of protoporphyrin dimethyl ester into its fluorovinyl analogs.

the solution in the presence of tetrakis(triphenylphosphine)-palladium to provide aryltrifluoroethenes (**22**) (see Scheme 3). First, we applied this method to 3,8-diiododeuterothiopyrroline dimethyl ester copper complex (**24**; met = Cu) obtained from deuteriothiopyrroline dimethyl ester (DTPDME: **23**), but it was quite difficult to control the temperature during the formation of the zinc reagent, and the solution of **21** was brown and turbid. The reaction of this solution with **24** under reflux yielded only 23% of 3,8-bis(trifluorovinyl)-DTPDME (**25**). A

similar reaction of 2-chloro-1,1-difluoroethene (**26**) gave a moderate yield (56%) of 3,8-bis(1-chloro-2,2-difluorovinyl)-DTPDME (**29**) (Scheme 3).

The poor yield of **25** seemed to be due to the low stability of trifluorovinyl lithium (**20**). Therefore, we decided to modify the procedure based on the idea that if zinc chloride were present when **20** was produced, it would react with the lithio derivative to form **21** before it had time to decompose. Thus, we added *sec*-butyllithium to a solution of **19** in the presence

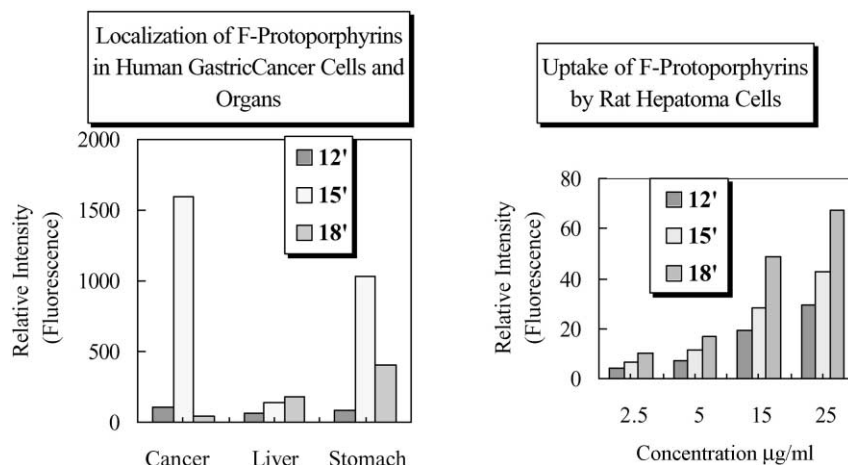
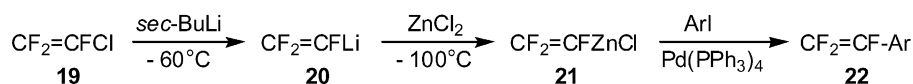
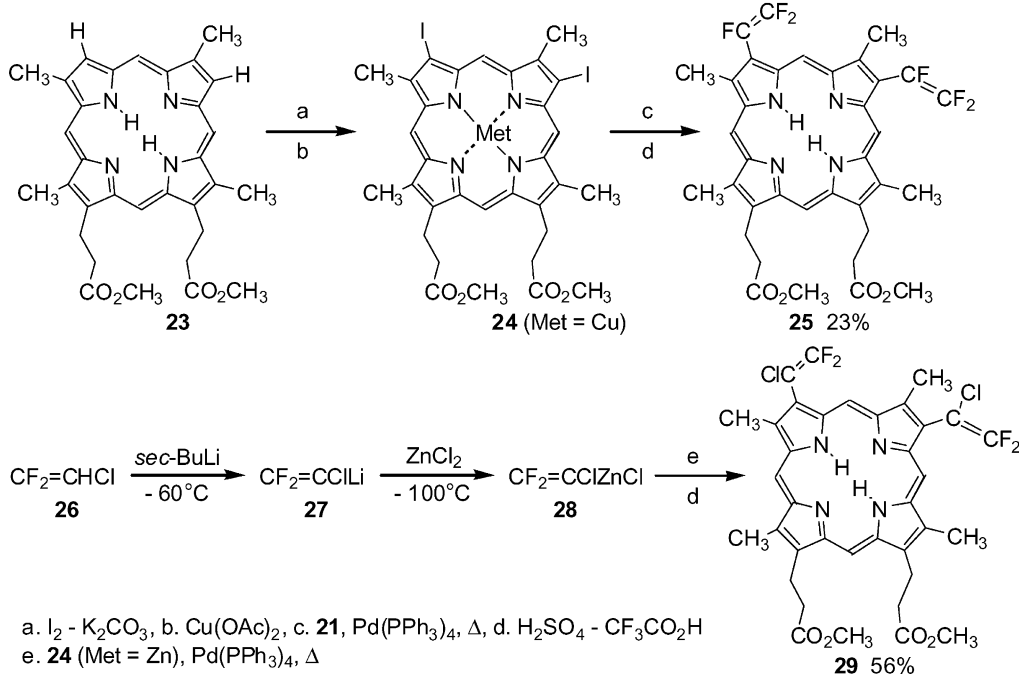
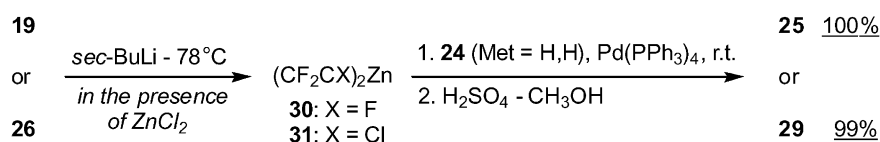


Fig. 2. Uptake of fluorinated porphyrins by cancer and cancer cells.

Normant's Procedure [19]*Application of Normant's procedure to diiododeuteroporphyrin dimethyl ester**Our modification of synthesis of fluorovinylzinc reagents and their application*

Scheme 3. Synthesis of fluorinated vinylzinc reagents and their applications Normant's procedure [22].

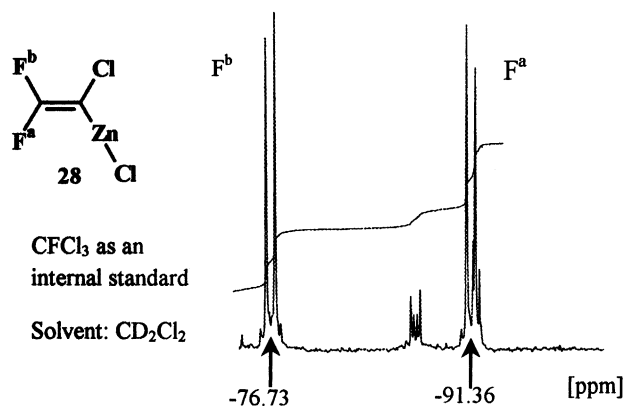


Fig. 3. ^{19}F -NMR spectrum of 1-chloro-2,2-difluorovinylzinc chloride.

of zinc chloride. The solution of zinc reagent thus formed was colorless and transparent, thus eliminating some technical difficulties. Interestingly, treatment of uncomplexed **24** (met = H, H) with this solution at room temperature gave a zinc complex of **25**, which was demetallated by treatment with sulfuric acid in methanol to give an excellent overall yield of **25**. By the same procedure as above, **26** gave **29** in 99% yield (see Scheme 3).

The differences between Normant's procedure and ours are remarkably large in terms of yields and reactivities: in the former case, the coupling reaction needed heating, but in the latter, it proceeded at room temperature. These results suggested some differences in the structures of reagents. Thus, we investigated the differences using ^{19}F -NMR. Our zinc reagent from **19** showed a more complex ^{19}F -NMR spectrum than expected from **21**.

Fortunately, we did obtain ^{19}F -NMR spectra of both forms of 1-chloro-2,2-difluorovinyl zinc reagent. A solution of **28** produced by Normant's procedure showed two doublets, as expected (see Fig. 3). The ^{19}F -NMR of the solution of the

zinc reagent from **26** by our procedure appears as three doublets, but integration could prove it to be two pairs of two doublets (Fig. 4). We tentatively assigned this as a mixture of *cisoid*- and *transoid*-bis(1-chloro-2,2-difluorovinyl)zinc (**31**), as shown in Fig. 4. Thus, one kind of fluorines of both isomers appears at a same chemical shift, and the other slightly different. We presume that our zinc reagent from **19** must be a similar mixture of bis(trifluorovinyl)zinc (**30**).

2.2.2. Synthesis of fluorinated analogs of protoporphyrin using bis(fluorovinyl)zinc reagents

2.2.2.1. Replacement of both vinyl groups.

The new simple methodology for introducing a trifluorovinyl or a 1-chloro-2,2-difluorovinyl groups into a porphyrin ring was used in conjunction with the Wittig methodology described in Section 2.1 for making difluorovinylporphyrins to effect the introduction of both types of groups into deuterioporphyrin dimethyl ester (**23**).

We had just succeeded in developing an efficient formylation of **23** [23] via its copper complex (**32**), which was treated with trimethyl orthoformate in the presence of tin(IV) chloride, followed by demetallation with sulfuric acid and trifluoroacetic acid, and separation by column chromatography gave 3- and 8-formyl compounds (**33** and **34**) in the yields of 46 and 47%, respectively (see Scheme 4). Heating **33** or **34** with sodium chlorodifluoroacetate and triphenylphosphine in NMP afforded difluorovinyl compounds (**35** or **36**), iodination of which in the presence of potassium carbonate gave the corresponding iodo compounds (**37** and **38**) in high yields. When these transformations were carried out in reverse order, we encountered with some difficulties. Thus, iodination of **33** or **34** gave formyl iodo compounds, but these were hardly soluble in any solvents and quite difficult to purify.

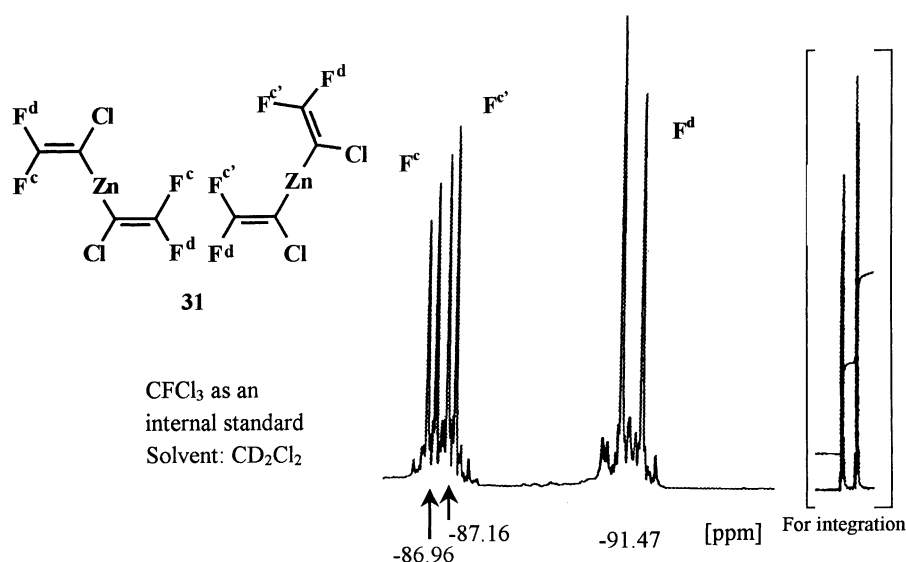
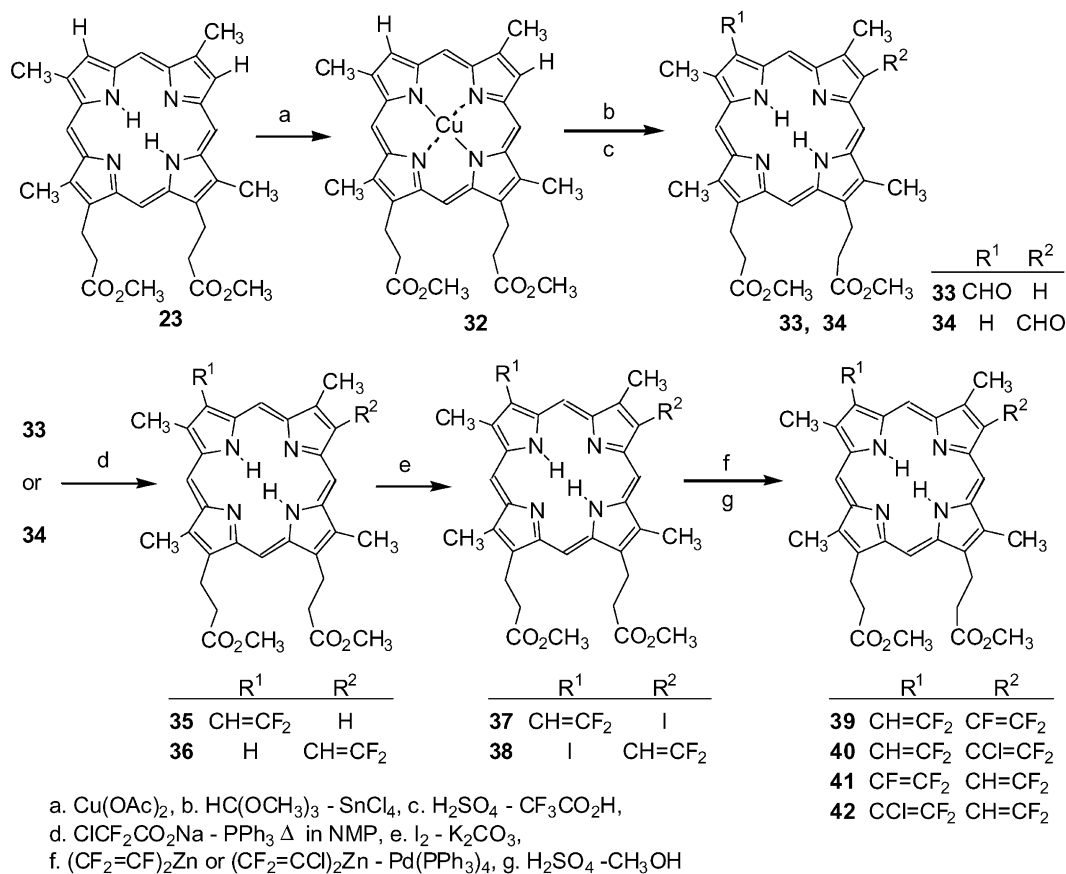


Fig. 4. ^{19}F -NMR spectrum of our zinc reagent (**31**) from 1,1-difluoro-2-chloroethene.



Scheme 4. Synthesis of fluorinated analogs of protoporphyrin with a difluorovinyl and a trihalovinyl groups [24].

Coupling reactions of (difluorovinyl)iodo-DPDME (**37** and **38**) with bis(trifluoro vinyl)zinc (**30**) or bis(1-chloro-2,2-difluorovinyl)zinc (**31**) in the presence of tetrakis(triphenylphosphine)palladium provided DPDME derivatives having a 2,2-difluorovinyl and a trifluorovinyl or a 1-chloro-2,2-difluorovinyl groups at the 3- or 8-position (**39–42**). All the reactions are shown in Scheme 4 [24].

These syntheses were accomplished recently, and investigation of the biological behaviors of the novel fluoroporphyrins acquired is now in progress.

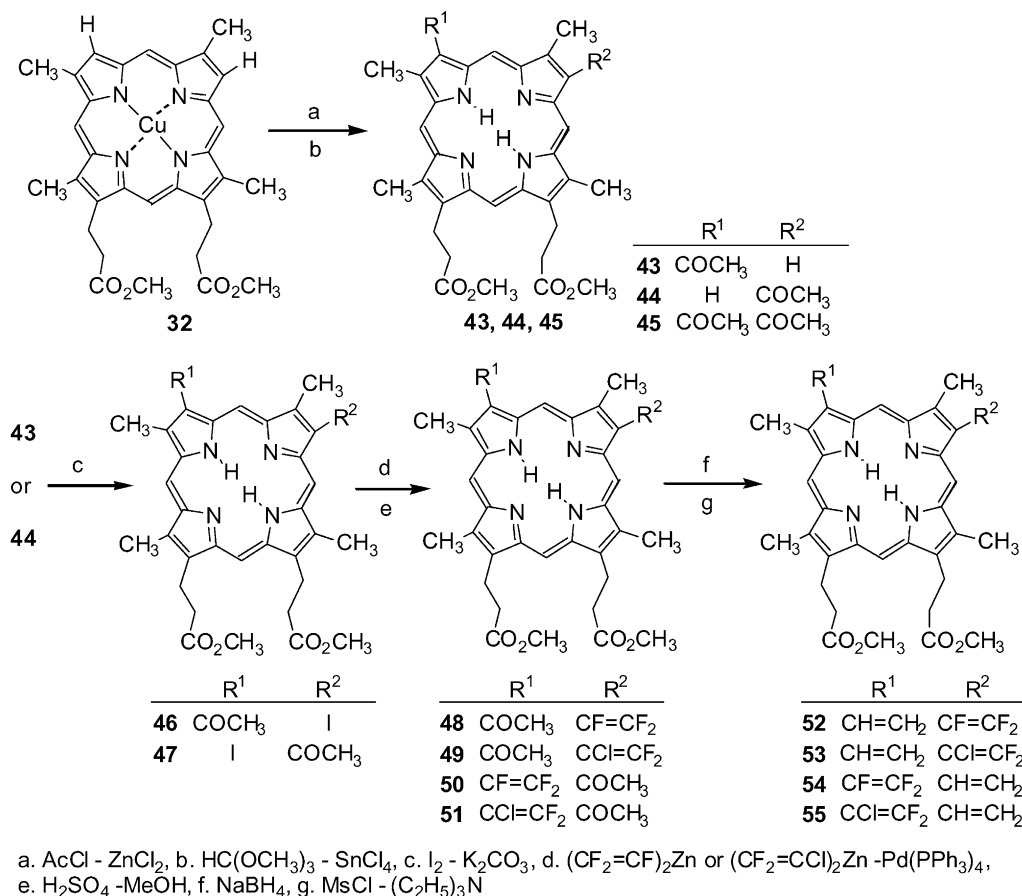
2.2.2.2. Replacement of one vinyl group. Our strategy here was to introduce into DPDME (**23**) a trifluorovinyl or a 1-chloro-2,2-difluorovinyl group at the 3- or 8-position using our bis(fluorovinyl)zinc reagents, having previously put in place at the 8- or 3-position an acetyl group, which, in the final step of the syntheses, could be converted to a vinyl group via reduction/dehydration sequence (Scheme 5).

In practice, the copper complex of DPDME (**32**) was treated with acetyl chloride in the presence of zinc chloride, followed by demetallation with sulfuric acid and trifluoroacetic acid, to give 3- and 8-monoacetyl-, and 3,8-diacetyl-DPDME (**43**, **44** and **45**) in yields of 27, 17 and 25%, respectively. Since unsubstituted vinyl groups at the 3- or 8-position of a porphyrin ring are highly reactive, the

conversion of the acetyl to vinyl group was left until a fluorovinyl moiety was in position. Thus, **43** and **44** were iodinated to give high yields (97 and 99%) of iodo compounds (**46** and **47**). These iodo compounds were then treated with bis(fluorovinyl)zinc reagents (**30** or **31**) to provide DPDME derivatives **48–51**. Reduction of the acetyl group of these compounds with sodium borohydride, followed by dehydration with methanesulfonyl chloride and triethylamine afforded four target compounds (**52–55**) in moderate to good yields (66, 61, 51, and 53%, respectively) [25]. Study on their biological features is now in progress.

3. Synthesis of fluorinated analogs of hematoporphyrin

Hematoporphyrin derivative (HpD), which can be used as a PDT photosensitizer, was derived from hematoporphyrin (**2**) by treatment with sulfuric acid in acetic acid, but this gives a complex mixture of several porphyrins, as mentioned before (see Section 1). So, we embarked on the synthesis of pure fluorinated analogs of hematoporphyrin with 2,2,2-trifluoro-1-hydroxyethyl (TFHE) in place of one or both 1-hydroxyethyl groups.



Scheme 5. Synthesis of fluorinated analogs of protoporphyrin with a trifluorovinyl and a chlorodifluorovinyl groups [22].

3.1. Trifluorohydroxyethylation of deuteroporphyrin

Our first attempt to introduce TFHE group(s) into deuteroporphyrin dimethyl ester (DPDME: **23**) was based on the reported synthesis of TFHE-benzene derivatives via the reaction of benzene or its derivatives with trifluoroacetaldehyde ethyl hemiacetal in the presence of a Lewis acid [26]. Several Lewis acids (FeCl_3 , TiCl_4 , SnCl_4 , SbCl_5 , ZnCl_2 and AlCl_3) were examined as a catalyst, but the yields of TFHE-DPDMEs (**56** and **57**) amounted to only 26% in total, and the starting material (**23**) was recovered in 66% when AlCl_3 was used. Other Lewis acids were less effective (see Scheme 6).

To improve the yields of TFHE-DPDMEs, we tried the Friedel–Crafts reaction of **23** with CF_3CHO . Reaction of **23** with CF_3CHO in the presence of AlCl_3 gave mono-TFHE compounds (**56** and **57**) and bis-TFHE **58** in 40–50% and a few percent yields, respectively. Thus, the yields were improved considerably. Further, addition of **23** to an AlCl_3 – CF_3CHO complex in CH_2Cl_2 formed at -78°C , followed by heating the mixture at 50°C for 4 h, gave mono-TFHE compounds **56** and **57** in 36 and 26%, respectively, and the bis-TFHE compound (**58**) in 6% yield, with 13% recovery of the starting material (**23**) (see Scheme 6).

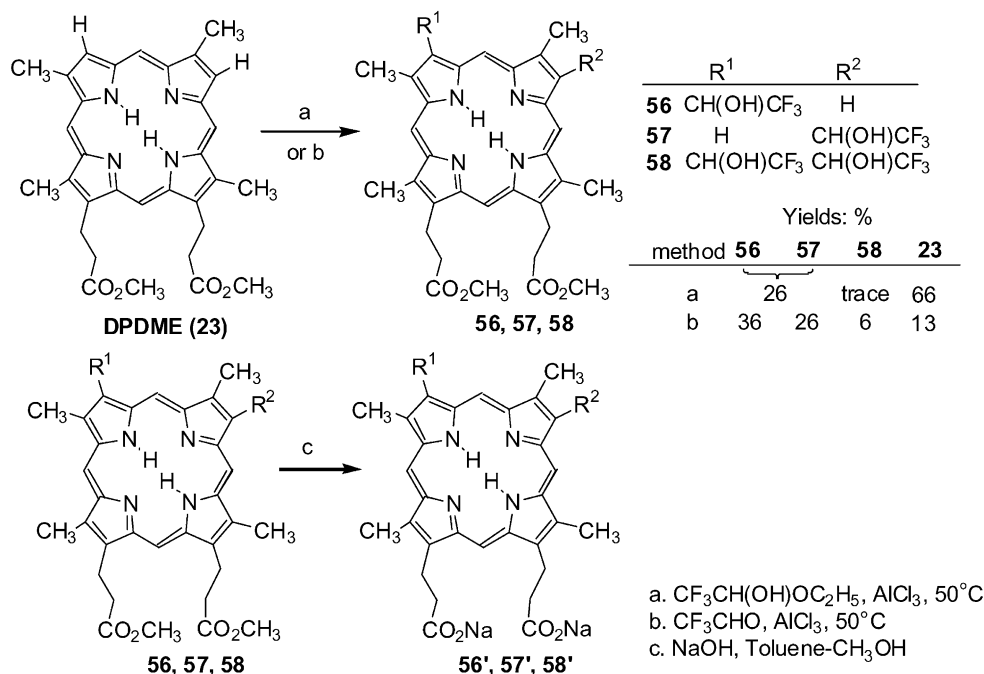
These porphyrin derivatives (**56–58**) were hydrolyzed to give Na salts (**56'–58'**), needed for use in uptake studies

involving human liver cancer cells (JTC-16). Each TFHE-DPN a (**56'–58'**) was added to the culture media of JTC-16 cells and incubated for 48 h; the cells were then washed with buffer solution and extracted with (*iso*- C_3H_7) $_2\text{NH}-\text{CH}_3\text{OH}$ [27]. The intensity of fluorescence of this extract was measured by fluorophotometry. The results, shown in Fig. 5 [28,29], reveal that 3,8-(TFHE) $_2$ -DPNa (**58'**) is taken up more effectively by human liver cancer cells than TFHE-DPNa **56'** or **57'**, and that of these mono-TFHE compounds, the 8-TFHE derivative (**57'**) is taken up more effectively than its 3-isomer (**56'**).

3.2. Synthesis of 'trifluorohematoporphyrins'

In the above investigation of the uptake of TFHE-containing porphyrins by cancer cells, 3,8-TFHE-DPN (**58'**), a 'hexafluorohematoporphyrin', was found to be taken up more readily than its 3- or 8-TFHE analogs. To clarify the effect of the fluorine substituents, we set out to synthesize 'trifluorohematoporphyrins', namely compounds carrying one TFHE and one hydroxyethyl group.

For this purpose, derivatization of the mono-TFHE-DPDMEs **56** and **57** to mono(1-hydroxyethyl)TFHE-DP, trifluoro analogs of HP, was examined. Thus, **56** and **57** were each converted to a Cu complex and then the hydroxyl



Scheme 6. Introduction of a trifluorohydroxyethyl groups into DPDME.

groups were protected by acetylation to give **59** and **60**. Treatment of these products with acetyl chloride in the presence of a Lewis acid, followed by demetallation, gave ring-acetylated products **61** or **62**. In this acetylation, the identity of the Lewis acid played an important role: tin(IV) chloride was effective for acetylation of **59**, but not for **60**, and titanium(IV) chloride was suitable for the acetylation of **60**. The acetyl groups of **61** and **62** were reduced with sodium borohydride and the *O*-acetyl groups were removed by sodium hydroxide to give the desired sodium salts (**61'** and **62'**). These reactions are shown in Scheme 7.

In a preliminary test of their uptake by human liver cancer cells, hexafluorohemato-porphyrin **58'** was found to be taken

up more readily than trifluorohematoporphyrins **61'** and **62'** [30].

3.3. Formylation of TFHE-DPDMEs and modification of the formyl groups

Looking for fluorinated porphyrins that would localize more selectively to a cancer, we decided to introduce further substituents into AcO-TFE-DPDME-Cu complexes (**59** and **60**). Thus, reaction of **59** and **60** with trimethyl orthoformate in the presence of CF₃COOH, followed by demetallation with CF₃COOH and H₂SO₄, was carried out to give formyl compounds **63** and **64** and hence, via hydrolysis with NaOH, sodium salts **65** and **66**. Next, we tried the reaction of these formyl compounds (**63** and **64**) with carbanion equivalents such as a Grignard reagent and an enolate ion. Reaction with vinylmagnesium bromide gave the corresponding 1-hydroxy-2-propenyl (HPE) compounds, which were hydrolyzed to sodium salts (**67** and **68**, respectively). Reaction of **63** and **64** with 2-trimethylsiloxyoctene in the presence of titanium(VI) chloride gave 3-oxo-1-nonenyl (ONE) compounds, both of which was hydrolyzed to sodium salts (**69** and **70**, respectively) [31]. These results are summarized in Scheme 8.

The localization of the six sodium salts (**65–70**) thus acquired in tumor tissue and organs was investigated. The results are shown in Fig. 6 [31]. These suggest that some fluorine-containing porphyrins would localize to cancer more readily than HpD. Uptakes of those porphyrins which have fluorine substituents at the 8-position, such as **66** and **68**, are especially remarkable. These results suggest that these porphyrins localize to cancer more readily than HpD, and/or localize to liver and kidney less readily than HpD.

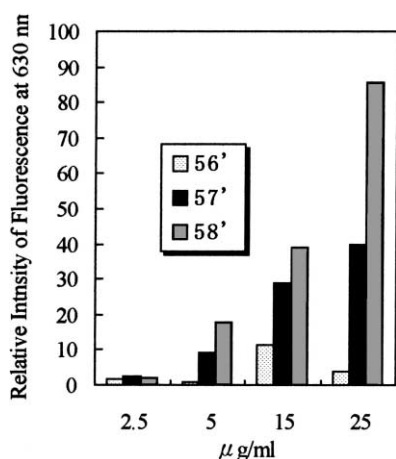
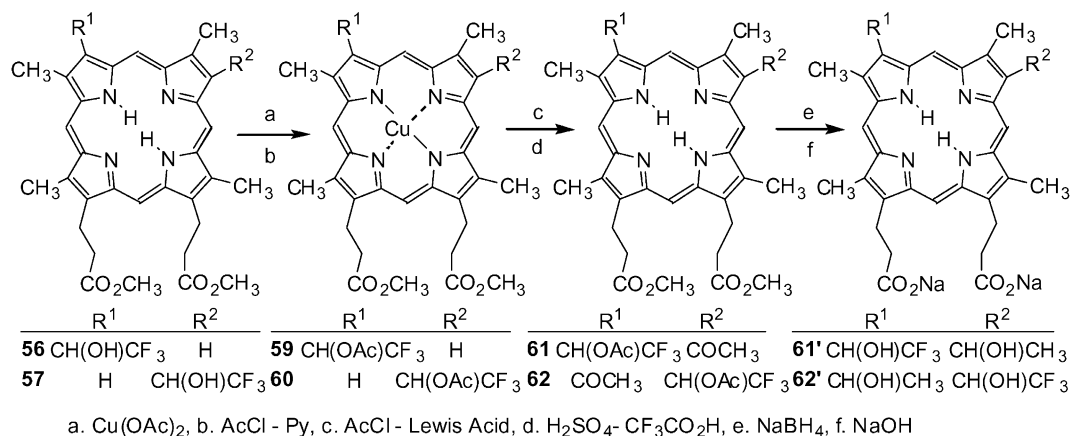
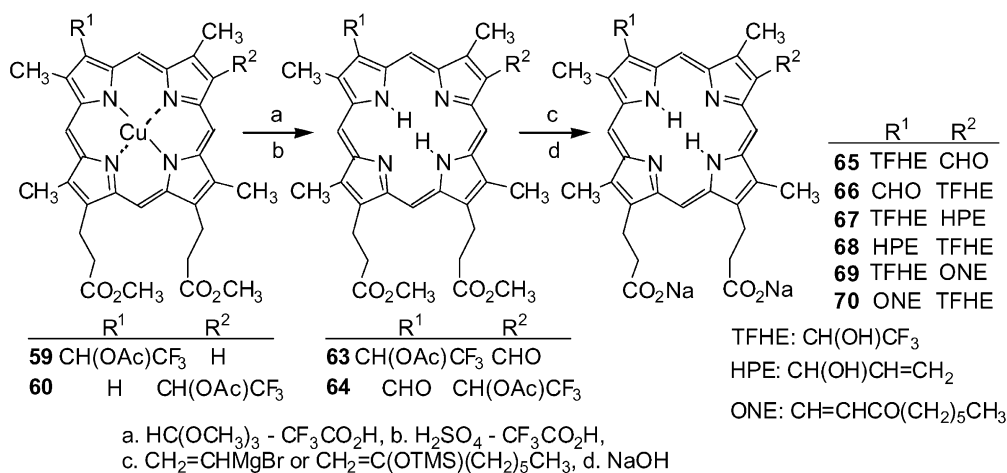


Fig. 5. Uptake of sodium salts of (trifluorohydroxyethyl)-deuteroporphyrins by JTC-16 (human liver cancer cells).



Scheme 7. Synthesis of trifluorohematoporphyrins.



Scheme 8. Formylation of TFHE-DPDMEs and modification of the formyl group.

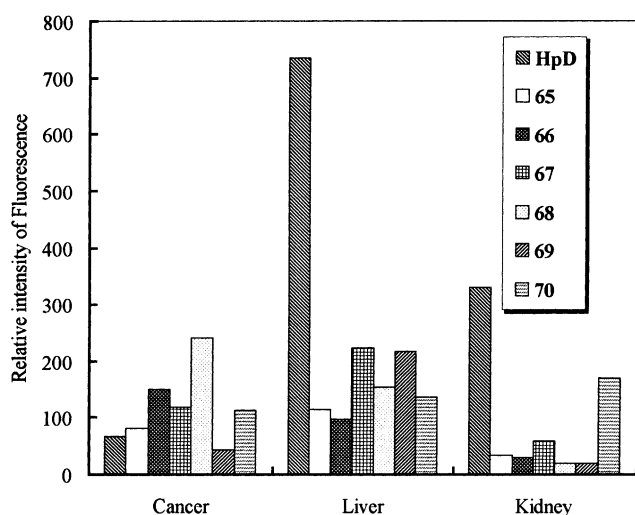


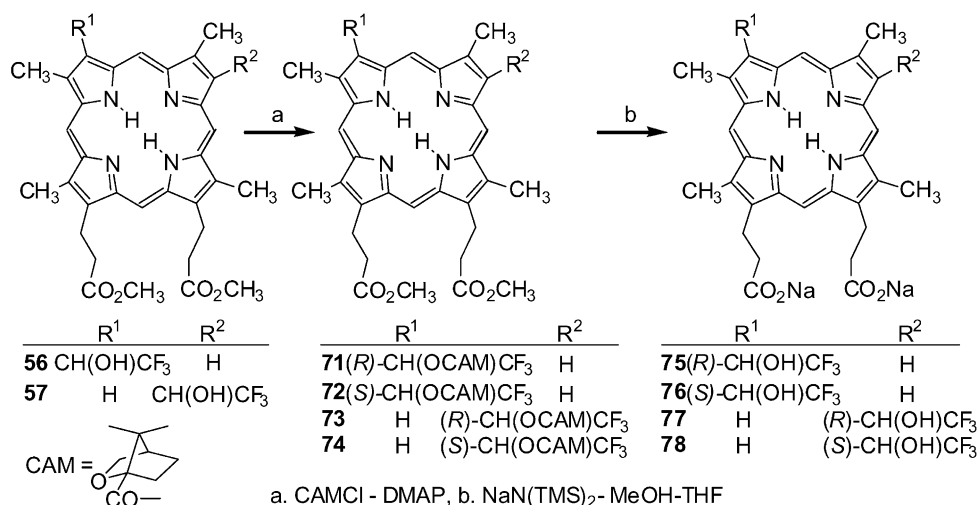
Fig. 6. Uptake of fluorinated porphyrins (65–70).

4. Synthesis of chiral fluorine analogs of hematoporphyrin

HP (2) has two chiral centers, located in the hydroxyethyl groups, but all commercial samples are mixtures of diastereomers. Generally speaking, biological activities of most chiral compounds depend on their chirality. Thus, the degree of uptake of chiral fluorinated analogs of HP by tumor could well depend on the chirality: to throw light on the situation, we planned to synthesize and test some chiral fluorinated analogs related to HP.

4.1. Optical resolution of mono-TFHE-deuteroporphyrin dimethyl esters

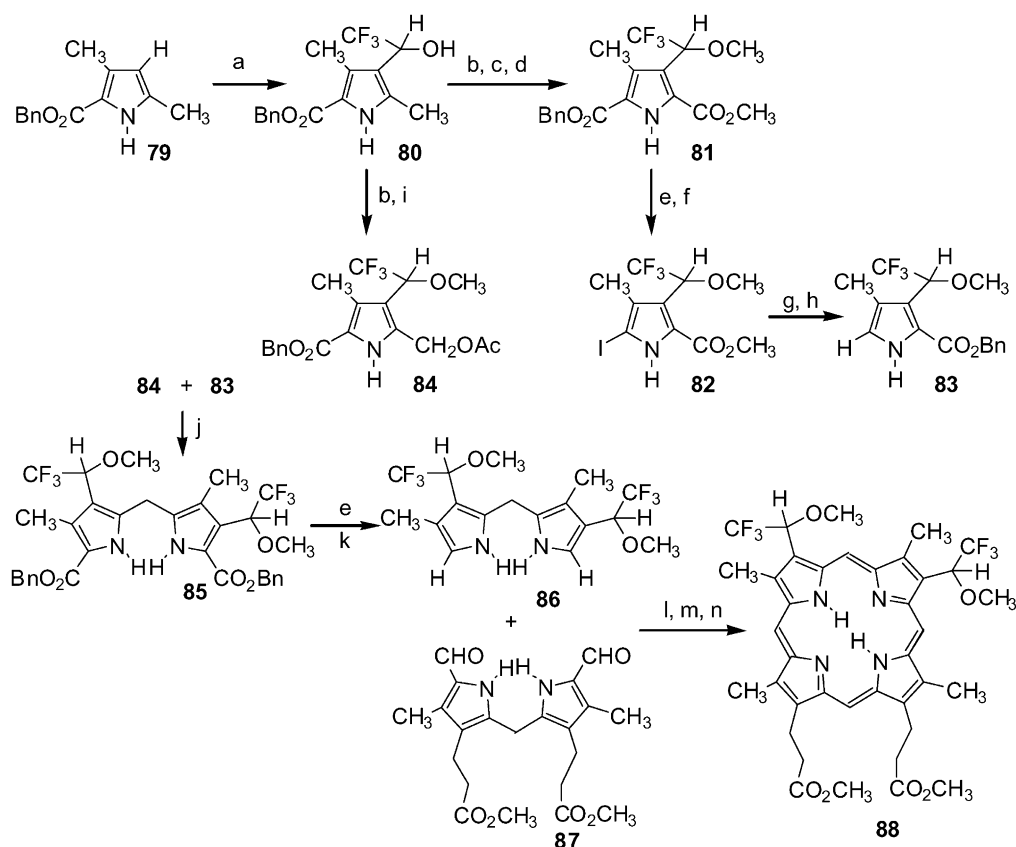
First, optical resolution of bis-TFHE-DPDME (58) was attempted derivatizing it to (*S*)-camphanic ester, but it contained four diastereoisomers, and we could not separate



Scheme 9. Optical resolution of mono-TFHE-DPDMEs.

a mixture of diastereomeric esters. Next, we tried optical resolution of mono-TFHE-DPMEs **56** and **57**, derivatizing them to (*S*)-camphanic esters. Diastereomeric esters of both isomers were separated efficiently using silica gel column chromatography. Separated esters (**71–74**) were then

hydrolyzed with sodium bis(trimethylsilyl)amide in MeOH-THF to sodium salts, 3-(*R*)-, 3-(*S*)-, 8-(*R*)-, and 8-(*S*)-TFHE-DPNa (**75–78**) (see Scheme 9) [32]. Each salt was subjected to a preliminary test for uptake by human gastric cancer cells. Interestingly, the 3-(*S*) compound **76** was taken



Scheme 10. Model synthesis of noon-chiral bis-TFHE-DPDME by ring-closure.

up fifteen times more than its 3-(*R*) isomer **75**, while the 8-(*R*) isomer **77** was a little more than 8-(*S*) **78** [32].

4.2. Model synthesis of TFHE-DP by ring closure

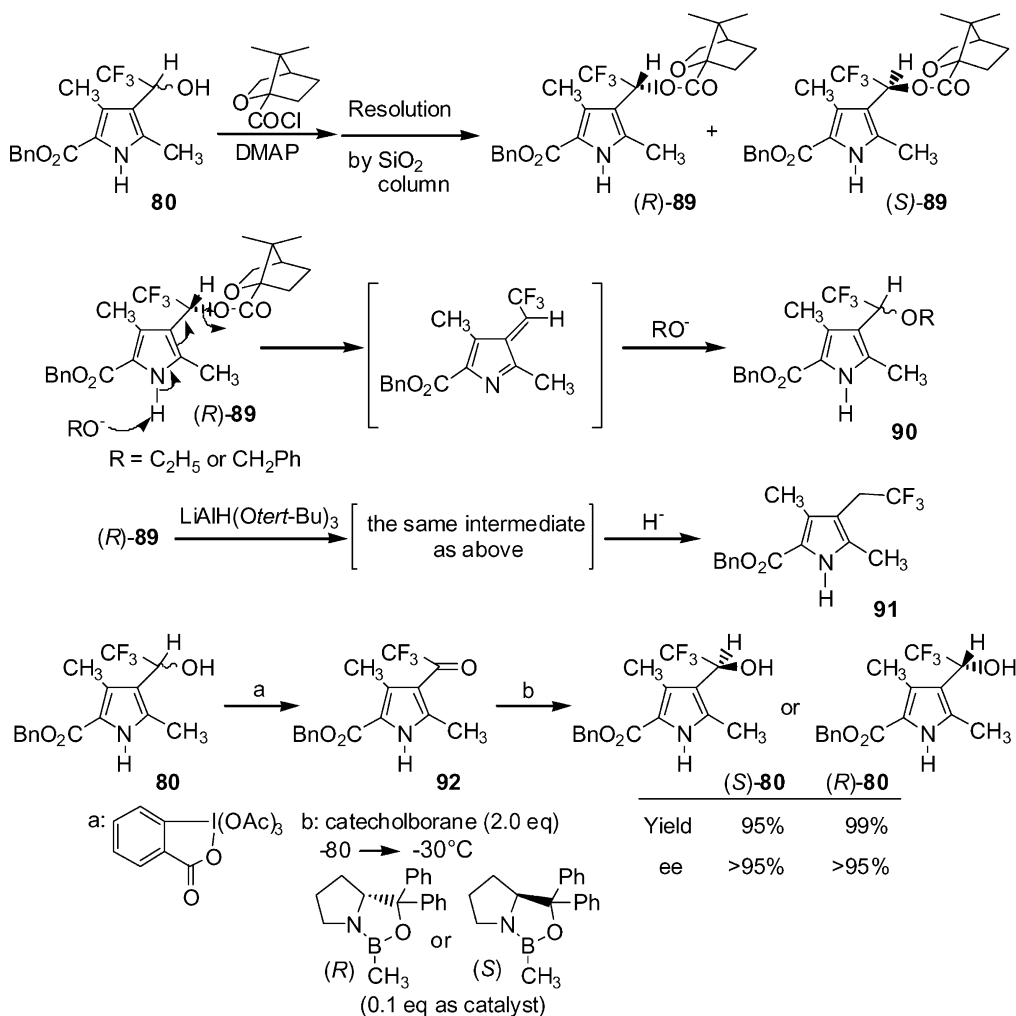
The above results suggested that chirality associated with a TFHE group on porphyrin can play an important role in the uptake of porphyrins by cancer cells. However, having failed to achieve any resolution of bis-TFHE-DPDME (**58**) (see above), we hoped to solve this problem via the total synthesis of chiral analogs of HP by ring-closure methodology, starting from a pyrrole derivative with a chiral TFHE group. First, we examined ring-closure using optically inactive pyrrole derivatives as model compounds.

Our strategy involved treatment of benzyl 3,5-dimethylpyrrole-2-carboxylate (**79**) with trifluoroacetaldehyde ethyl hemiacetal in the presence of a Lewis acid to give a 4-TFHE derivative (**80**), which was derivatized to the rings A (**84**) and B (**83**) of the porphyrin system through several steps, as shown in Scheme 10 [33,34]. Initially, the hydroxyl group of the TFHE group in racemic **80** was protected by conversion

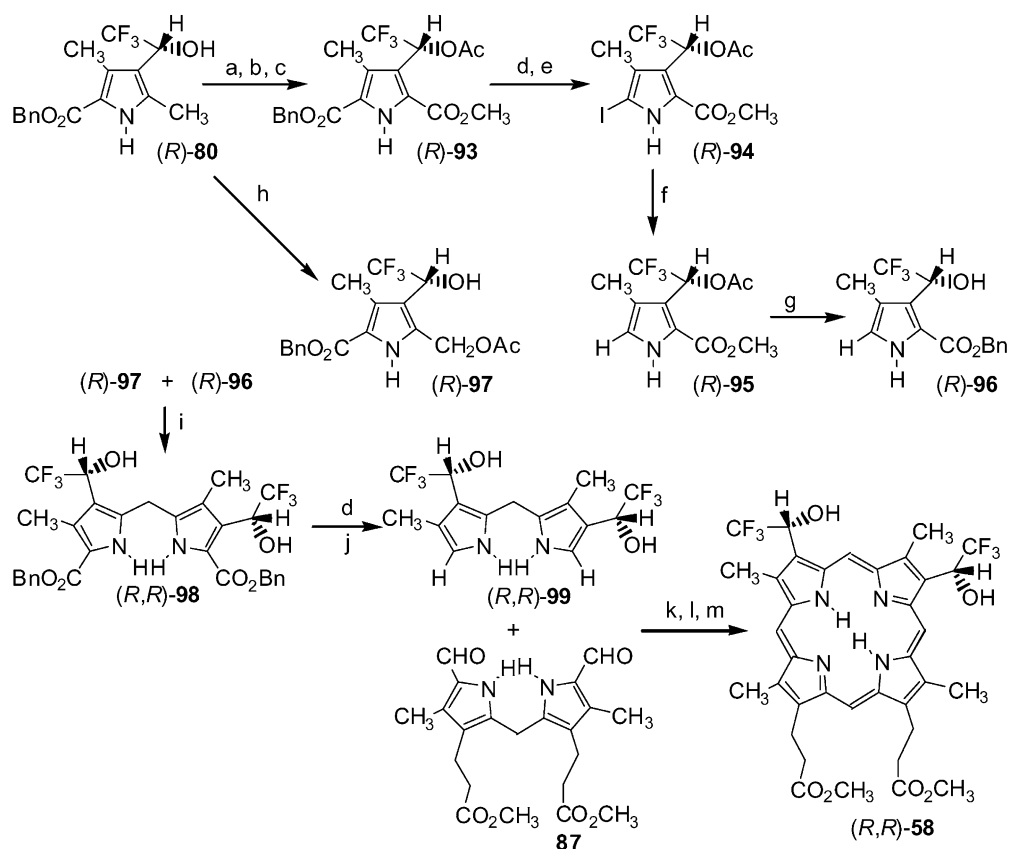
to OMe, then the 5-methyl group was manipulated to give methyl ester **81**. Hydrogenolysis of the benzyl ester at the 2-position, and then *ipso*-iodination with simultaneous decarboxylation afforded the 2-iodo compound **82**, which was reduced to provide the 2-hydrogen compound (**83**). After *O*-methylation of **80**, the 5-methyl group was partially oxidized with lead tetraacetate to give acetoxymethyl compound **84**. Condensation of **83** and **84** catalyzed by boron trifluoride etherate gave the bis(pyrrole)methane **85**. The benzyl ester groups of **85** were hydrogenolyzed, and the resulting carboxyl groups were removed thermally to give **86**; MacDonald condensation of this product with **87** (obtained according to the literature) gave *O,O*-dimethyl the analog (**88**) of **58** [33,34].

4.3. Optical resolution and synthesis of pyrroles carrying a chiral TFHE groups (see Scheme 11)

Having succeeded in synthesizing a fluorinated analog of HP by ring closure, we needed to acquire a chiral form of the TFHE-pyrrole **80** for conversion to a chiral bis-TFHE-DP, hexafluoro analog of HP.



Scheme 11. Synthesis and resolution of optically active TFHE-pyrroles [35].



a. AcCl - DMAP, b. SO_2Cl_2 then H_2O , c. CH_2N_2 , d. $\text{H}_2/\text{Pd-C}$, e. I_2 - NaI, f. H_2/PtO_2 , g. BnOH - base, h. $\text{Pb}(\text{OAc})_4$, i. $\text{BF}_3\text{-OEt}_2$, j. base - Δ , k. TsOH , l. $\text{O}_2/\text{Zn}^{2+}$, m. H_2SO_4

Scheme 12. Synthesis of chiral bis-TFHE-DPDMEs using the chiral pyrrole derivatives **80** [36].

Initially, **80** was converted to diastereomeric (*S*)-camphanic esters (**89**), and the mixture was successfully resolved by silica gel column chromatography (>95% de). The absolute configuration of one ester (*S*)-**89** was determined by X-ray analysis [35].

Unfortunately, however, hydrolysis of chiral **89** with sodium ethoxide or benzyloxide resulted in the formation of racemic ethers (**90**) of **80** (see Scheme 11 for a plausible mechanism). Attempted reductive deacylation of (*R*)-**89** with lithium Tris-*tert*-butoxy-aluminumhydride gave a deacyloxylation product (**91**), probably through the same intermediate involved in the formation of **90**.

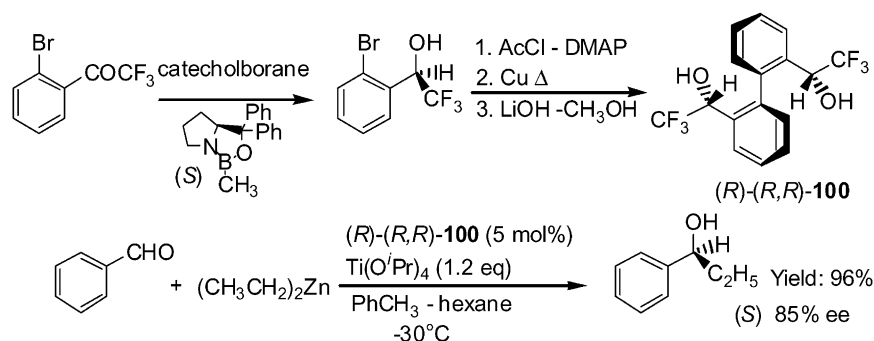
Finally, racemic **80** was oxidized with the Dess–Martin reagent to trifluoroacetyl-pyrrole **92**, followed by its asymmetric reduction with catecholborane in the presence of Corey's oxazaborolidine catalyst to give high yields of (*R*)- and (*S*)-**80** in excellent enantiomeric excesses. All the reactions discussed in this section are shown in Scheme 11 [35].

4.4. Ring formation using chiral TFHE-pyrroles [36]

Now that we had chiral TFHE-pyrrole compounds in hands, the ring-forming reactions described in Section 4.2

were followed with a few modifications. (*R*)-**80**, for example (see Scheme 12) was acetylated and treated with sulfonyl chloride followed by hydrolysis to give a carboxylic acid, which was converted with diazomethane to methyl ester (*R*)-**93**. The benzyl ester at the 2-position was then hydrogenolyzed and the 2-carboxylic acid was transformed to iodo compound **94**. This iodide was hydrogenolyzed to (*R*)-**95**, which was treated with sodium bis(trimethylsilyl)amide in benzyl alcohol to provide the 5-carboxylic benzyl ester (*R*)-**96** with simultaneous deacetylation of the TFHE group. Through these transformations, decrease of chirality was negligible. An especially interesting point is that hydrolysis of **95** proceeded without loss of the chirality, whereas the same reaction of chiral **89** had resulted in complete racemization (see Scheme 11). It seems that the electron-donating effect of the pyrrole ring may be cancelled by the electron-withdrawing effect of the 5-carboxylic ester group.

During these transformations the configurational integrity of the TFHE group was maintained, thus subsequent transformations in Scheme 12 were carried out without protection of its hydroxy group. Partial oxidation of the methyl group of (*R*)-**80** with lead tetraacetate gave the 5-acetoxymethyl compound (*R*)-**97** in high enantiomeric



Scheme 13. Synthesis and application of an axially chiral ligand.

excess. Condensation of (*R*)-**96** and (*R*)-**97** in the presence of boron trifluoride etherate gave dipyrrolemethane compound (*R,R*)-**98** in 88% yield and greater than 95% diastereomeric excess. Removal of the benzyl groups and decarboxylation gave (*R,R*)-**99**, condensation of which with **87**, followed by oxidation and demetallation, provided (*R,R*)-**58** (37% yield) in a high diastereomeric excess (85% de) (see Scheme 12).

Following the similar reactions from (*S*)-**80**, and combining (*R*)- or (*S*)-**96** with (*R*)- or (*S*)-**97**, we could obtain other three diastereoisomers, (*R,S*)-, (*S,R*)- and (*S,S*)-**58** (29, 41, and 25% yields, respectively) in high diastereomeric excesses (91, 89, and 86% de, respectively) [36].

All four diastereoisomers of **58** were hydrolyzed to sodium salts. Detailed study of the biological behaviors of these hexafluoroanalogs of HP are still in progress. Preliminary tests have shown that the (*S,R*)-isomer is the most effective, where take-up by cancer cells is concerned. This suggests again that chiral centers present in HpD must play an important role in biological activity.

Note that the very high enantioselectivity in the reduction of a trifluoroacetyl group with catecholborane in the presence of Corey's oxazaborolidine catalyst is quite useful for the synthesis of chiral 2,2,2-trifluoro-1-hydroxyethyl compounds (see Scheme 11). Using this methodology, we have synthesized an axially dissymmetric ligand with two chiral TFHE groups (**100**) (see Scheme 13). Ethylation of benzaldehyde with diethylzinc catalyzed by titanium tetraisopropoxide in the presence of **100** gave a quantitative yield of 1-phenyl-1-propanol in 85% de (see Scheme 13) [37].

This result means that **100** is a more effective ligand than BINOL in this reaction. Other applications of this ligand are now under investigation.

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